

SQA Release *Standards.*

If the basics fail, the release is denied. A field guide to what every developer must check before sending a build to QA.

The same basic mistakes. Every sprint.

These defects show up in QA again and again, across every category. They are not edge cases — they are the basics.

01

Empty fields submit

Login button stays clickable; form posts with blank email or password.

02

Plain-text password

Password rendered as text — visible while typing, in DOM, in payload.

03

Generic error messages

'Something went wrong.' — no field, no reason, no recovery hint.

04

Typos in error text

'Plese enter a vaild email' — visible to end users in production.

05

Mobile field accepts letters

Phone field allows 'abc-123'; saved without country code.

06

Upload accepts anything

No size cap, no type filter, no progress, no cancel.

07

Sticky header overlaps

Header covers content on scroll; first paragraph hidden.

08

Dropdown values cut off

Long selected value hidden because the field is too narrow.

09

Delete with no confirmation

Single click → record gone. No confirm. No undo. No log.

10

Reset link never expires

Forgot-password token reusable forever; sent over plain HTTP.

11

Report PDF cuts off mid-page

Numbers without commas, no page numbers, broken filename.

12

Stale copyright in footer

'© 2022' on a 2026 page; some footer links return 404.

THE NEW STANDARD

If the basics fail, the release is denied.

01

Self-check first

Developers run the basic checklist before opening a PR or asking for QA.

02

Same-day rejection

If a release contains basic defects, SQA returns it the same day — no triage cycle.

03

Re-submit clean

Build is retested only after the developer confirms the basics pass.

Twenty-five basics — checked on every build

01	Register	02	Login	03	Email	04	Password
05	OTP	06	Captcha	07	Forgot password	08	Buttons
09	Delete button	10	Error messages	11	Input field	12	Mobile number
13	Searching	14	Dropdown	15	Checkbox	16	Radio
17	URL field	18	Text area	19	Currency/Amount	20	Tooltip
21	Links	22	Upload	23	Download/Report	24	Header
25	Footer						

Registration form

Whatever you allow at sign-up, you live with forever. Weak rules now mean weak accounts later.

If you ship this — release is denied

- No password complexity — "123456" accepted
- Confirm-password mismatch only checked on Submit, not on blur
- Plus-alias trick (user+1@gmail.com) bypasses duplicate-account check
- No CAPTCHA — registration endpoint is open to bots
- No email verification step; pro features accessible immediately
- Spelling errors on the page: 'Submitt', 'Reigster'

Pass these and submit with confidence

- Password rules enforced: 8–16 chars, uppercase, number, special character
- Confirm-password match checked on blur, not just on Submit
- Plus-alias addresses normalized for duplicate detection
- CAPTCHA on registration; bot signups blocked
- Email verification required before pro features unlock
- Entered values retained after a validation error; spelling reviewed

Login

The most-used screen in any product. Mistakes here are visible to every user, every day.

If you ship this — release is denied

- Login button enabled when email or password is empty
- Wrong credentials → infinite spinner, no error message
- Password visible in plain text in the input field
- No lockout after 10+ failed attempts (brute force open)
- Pressing Enter does not submit the form
- Logged-in URL works in incognito after logout (token still valid)

Pass these and submit with confidence

- Login button disabled until both fields contain values
- Field-specific error: 'Email is required' / 'Invalid format'
- Password masked by default; eye icon toggles visibility
- Rate-limit / CAPTCHA after defined failed attempts
- Enter key and button click both submit the form
- Session token invalidated on logout; HTTPS enforced

Email field

Every form has one. Every form gets it wrong in the same handful of ways.



If you ship this — release is denied

- 'user@example.com' rejected because of leading/trailing spaces
- 'User@Example.com' and 'user@example.com' saved as different users
- 'user@example' accepted as valid (no TLD check)
- Generic 'Invalid input' error with no guidance
- No mobile keyboard hint — user types @ on alphabet layout
- SQL/script payloads pass through to the server unsanitized



Pass these and submit with confidence

- Whitespace auto-trimmed before validation and storage
- Email lowercased on save; comparison is case-insensitive
- Reject formats missing @, dot in domain, or with double dots
- Field-specific error: 'Please enter a valid email address'
- type="email" with autocomplete="email" / inputmode="email"
- Server-side sanitization for SQL injection and XSS

Password field

Security and usability live in the same input. Both must be solid before release.

If you ship this — release is denied

- Field uses type="text" — password visible while typing
- Password sent in plaintext in the network request body
- Password value visible in localStorage, cookies, or URL
- No strength indicator on registration / change-password
- Special characters (P@ss!w0rd#) silently rejected
- No paste support — password managers can't fill the field

Pass these and submit with confidence

- Field uses type="password"; eye icon toggles visibility
- Transmitted only over HTTPS; never visible in DevTools payload
- Never stored in localStorage / cookies in plain form
- Strength meter and rules visible during registration
- All printable special characters supported
- Paste (Ctrl+V and right-click) works for password managers

One-time password

A weak OTP flow defeats the entire purpose of two-factor authentication.

If you ship this — release is denied

- OTP length inconsistent — sometimes 4, sometimes 6
- OTP sent in URL query string — appears in logs and history
- OTP logged in plaintext on the server
- No expiry — same code accepted hours later
- Reusable — same code can verify multiple times
- No rate limit on verification — brute force succeeds in minutes

Pass these and submit with confidence

- OTP length is always 6 digits — never less, never more
- Transmitted in HTTPS POST body — never in URL or logs
- Server logs only request ID and outcome, never the OTP
- OTP expires after 5 minutes; expired codes always rejected
- Single-use — second attempt with same code returns 'already used'
- After 5 failed attempts the OTP is invalidated; resend has cooldown

Bot protection (CAPTCHA)

When CAPTCHA is broken, every protected endpoint is open. It must work on every browser.

✗ If you ship this — release is denied

- Fails silently when ad blockers are enabled — no fallback
- No audio CAPTCHA option — fails accessibility (WCAG 2.1)
- Verification happens client-side only — DOM manipulation bypasses it
- Same challenge served on every page reload
- No expiry — token from 30 minutes ago still works
- On mobile, the challenge popup is cut off and unusable

✓ Pass these and submit with confidence

- Server-side verification on every submit — not client-only
- Token is single-use; replay attempts rejected
- New challenge generated on every reload and after every failure
- Token expires after 2 minutes; expired tokens trigger refresh
- Audio fallback present; visual contrast OK for color-blind users
- IP rate-limit kicks in after repeated failures; mobile-tested

Forgot password

A password reset link is an authentication bypass if it leaks. Treat it like one.

If you ship this — release is denied

- Reset link uses http:// — token can be intercepted
- Token has no expiry; valid forever once issued
- Token is reusable — clicking the link twice still works
- Error reveals whether an email is registered (account enumeration)
- Old password keeps working after reset; other sessions stay alive
- No CAPTCHA / rate limit — endpoint can be abused for spam

Pass these and submit with confidence

- Reset link uses https:// only
- Token is single-use and expires within 15–30 minutes
- Same neutral message for valid and invalid emails — no enumeration
- On reset, all active sessions on other devices are logged out
- Old password rejected immediately; CAPTCHA + IP rate limit enabled
- Email arrives in inbox (DKIM/SPF/DMARC), not spam folder

Buttons & actions

A button that lies about its state — enabled when it shouldn't be — corrupts data.



If you ship this — release is denied

- Submit button clickable while form is invalid
- Double-click submits the form twice → duplicate records
- No loading state — user clicks again thinking nothing happened
- Only the text is clickable, not the surrounding button area
- Delete buttons fire immediately with no confirmation dialog
- Button text says 'OK' or 'Click here' instead of the action



Pass these and submit with confidence

- Disabled (greyed) while required fields are empty/invalid
- Disabled on click + spinner shown until response returns
- Loading indicator visible during async work
- Whole button area is clickable, with hover state on the area
- Destructive actions show a confirm dialog naming the item
- Text describes the action: 'Save changes', 'Delete user'

Delete (destructive actions)

Every "Are you sure?" you skip is a support ticket waiting to happen — sometimes a lawsuit.

If you ship this — release is denied

- Single click deletes immediately — no confirmation
- No undo, no recovery — destructive and silent
- Hard delete; no soft-delete flag, no audit log
- No RBAC check — any user can delete any record
- Cascade rules undefined — child rows orphan in the database
- No success message — user is unsure if anything happened

Pass these and submit with confidence

- Confirmation modal names the record and the consequences explicitly
- Cancel + Confirm clearly distinguished; destructive action is non-default
- Esc dismisses; modal is keyboard-accessible
- Soft delete by default; restorable within a documented window
- Undo toast appears for ~8 seconds after deletion
- RBAC enforced server-side; audit log captures user, timestamp, IP, record

Error messages

If your error message has a typo, the user assumes the whole product has typos.



If you ship this — release is denied

- 'Plese entr valid emial' — spelling/grammar mistakes
- Server stack trace shown in production UI
- Message disappears in 1 second before the user can read it
- Position shifts the form layout when the error appears
- Same generic 'Error occurred' used for every field
- No 500 page — blank white screen on server error



Pass these and submit with confidence

- Spelling and grammar reviewed before merge
- User-friendly server errors, with a request/error ID
- Error stays visible until corrected or the user dismisses it
- Reserved space prevents layout shift when error appears
- Field-specific text: 'Mobile number must be 11 digits'
- Branded 500 / 404 pages with a recovery CTA

Input field

Almost every form is a stack of single-line inputs. Validation, sanitization, character handling — get any of them wrong and the whole form is unsafe.

✗ If you ship this — release is denied

- Empty field accepted; required-field validation never runs
- Leading and trailing spaces stored as-is — 'Iqbal ' ≠ 'Iqbal' in lookups
- HTML rendered raw on read view — XSS open; SQL injection unsanitized
- No max-length cap — 50,000-character paste breaks the database
- Bangla / emoji become ??? (latin1 charset, not utf8mb4)
- Validation fires only on Submit — user finds errors at the bottom

✓ Pass these and submit with confidence

- Required validation on blur AND submit, with field-specific error
- Whitespace trimmed; inner multi-spaces normalized to one
- HTML escaped on render (XSS-safe); SQL parameterized server-side
- Max-length enforced at field + DB level; live counter visible
- utf8mb4 charset — Bangla, emoji, mixed-language preserved
- Disabled / readonly states visually distinct; standard shortcuts work

Mobile number field

Mobile is a regional, format-sensitive identifier. Treat it like one — not like a free-text field.



If you ship this — release is denied

- Field accepts letters and special characters
- Saved without country code; cannot be searched cross-region
- No length validation — accepts 6-digit and 25-digit values
- Whitespace and dashes break the lookup query downstream
- Country dropdown missing — only Bangladeshi numbers handled
- Same number storable as 01711... and +8801711... — duplicates in DB



Pass these and submit with confidence

- Field accepts digits only; length matches the selected country
- Country code prefix visible (with flag); BD default for Bangladesh
- Stored in E.164 international format (+8801711234567)
- Whitespace, dashes and brackets normalized server-side
- Validation message specific to the country (BD: 11 digits, starts with 01)
- Duplicate detection works across all input formats of the same number

Search field

Users judge the entire product by how the search behaves. Placeholder, results, empty state — all of it must be designed.

If you ship this — release is denied

- No search icon, no placeholder text — looks like any other input
- Pressing Enter does not trigger search — must click button
- Empty state shows just 'Wrong' — no helpful message, no suggestions
- No loader while results fetch — page appears frozen
- XSS not sanitized — `<script>` in query renders on results page
- Pagination breaks past page 3

Pass these and submit with confidence

- Magnifying-glass icon visible; placeholder reads 'Search providers, routes...'
- Enter and Search button both trigger the query
- Suggestions appear as user types (autocomplete with highlighted match)
- Loader appears if results take > 200 ms
- Empty state shows 'No results found' + related keywords
- XSS sanitized; SQL parameterized; pagination tested through last page

Dropdown / select

Every dropdown is data: ordered, complete, scrollable. Skip the basics and the form silently breaks.

✗ If you ship this — release is denied

- Misspelled value ('Bangaldesh') selected by default
- Casing inconsistent across options ('srilanka' vs 'Pakistan')
- Blank/empty option in the middle of the list
- Duplicate entries in the same list
- Order looks random — neither alphabetical nor by spec
- Dropdown does not open via keyboard or first-letter typing
- Long selected value cut off because the field is too narrow — only first few characters visible, with no indication that more text exists

✓ Pass these and submit with confidence

- All values present per the spec — none missing or extra; spell-checked
- Consistent Title Case across all options
- No blank options (except an explicit 'Select' placeholder if intended)
- No duplicates; order matches the spec
- Long values truncate cleanly with ellipsis; built-in search for long lists
- Tab-able, arrow-key navigable, type-to-jump enabled; cross-browser parity
- Long selected values truncate with ellipsis and show full text on hover, OR the field expands to fit — never silently hidden

Checkbox

Misalignment, missing labels, broken keyboard support — all of these silently produce wrong data.

If you ship this — release is denied

- Label vertically misaligned with the checkbox
- Clicking the label does not toggle the checkbox (no for/id link)
- Spelling errors in label text ('trem & conditons')
- Checkbox with no visible label — only the developer knows what it does
- Duplicate label texts in the same group
- No keyboard support — Space key does not toggle; no focus ring

Pass these and submit with confidence

- Each checkbox has a clear, distinct label linked via for/id
- Clicking the label toggles the checkbox
- Spelling reviewed; no duplicates; order matches the spec
- Visible focus ring when tabbed; Space toggles the focused checkbox
- Touch target $\geq 44 \times 44$ px on mobile
- Selection retained after a form validation error — user does not re-tick

Radio button

Radio buttons promise mutual exclusion. Break that promise and the form is meaningless.

If you ship this — release is denied

- Different name attributes — multiple options selected at once
- Spelling errors ('Enterprise') in label text
- Order does not match the spec
- Clicking the label text does not select the radio
- No visible focus indicator when navigated by keyboard
- Selection lost on validation error — user has to re-pick

Pass these and submit with confidence

- All radios share the same name — only one selectable at a time
- Order matches the spec; clicking the label selects the radio
- Visible focus indicator when group receives keyboard focus
- Tab moves into group; arrow keys switch within group
- Mandatory group: clear 'Please select an option.' on empty submit
- Selection retained after validation error; touch $\geq 44 \times 44$ px

URL input

A URL field is a parser, not a textbox. If it accepts garbage, the database stores garbage.

✗ If you ship this — release is denied

- Strings with no dot accepted as a URL ('example' saved as-is)
- No protocol enforcement — bare domains saved without https://
- Leading and trailing whitespace not trimmed
- javascript: URIs accepted — XSS waiting to happen
- Encoded URLs and query parameters break
- Generic 'Wrong URL' error — no guidance on what is wrong

✓ Pass these and submit with confidence

- Both http:// and https:// accepted; bare domains auto-prepend https://
- Leading/trailing whitespace trimmed automatically
- Unsafe schemes (javascript:, data:, file:) rejected
- Encoded URLs, query parameters, anchor fragments preserved
- Max length enforced (e.g. 2048 chars) with a clear error
- Specific error messages per case; paste works in all major browsers

Text area (multiline input)

Anywhere users can paste, attackers can paste payloads. Sanitize, count, and constrain — every time.

If you ship this — release is denied

- `<script>` renders and executes on the result page — XSS
- No SQL parameterization — single-quote injection works
- No max-length cap — 1 MB paste accepted, breaks the database
- No word/character count visible to the user
- Drag-and-drop file paste accepts .exe and .js without warning
- Validation error closes the field — user loses their input

Pass these and submit with confidence

- HTML and JavaScript stripped or escaped server-side
- SQL parameterized; injection attempts fail silently
- Hard char cap with live counter and 'X / max' indicator
- Word count updates on type, paste, and right-click paste
- Drag-and-drop only accepts whitelisted text formats
- Validation error keeps the field editable; content not lost

Currency / amount field

A number field that accepts free text creates messy data, broken reports, and angry finance teams.

❌ If you ship this — release is denied

- Multiple decimal points accepted — value reads '1,000,000.5.5'
- Leading comma and letters allowed in the field
- No live formatting — user types 1000000, sees 1000000
- Saved unformatted in the DB; reports show '1000000' with no commas
- Negative values handled inconsistently — minus sign sometimes stripped
- Maximum allowed value not enforced — overflow on aggregations

✅ Pass these and submit with confidence

- Commas inserted automatically as user types: 1000 → 1,000 → 10,000
- Decimal allowed once; subsequent decimals rejected
- Negative values supported; minus sign always preceded by a digit
- Letters and special characters rejected with specific messages
- Stored as raw number in DB; commas added in display layer only
- Same formatting in every view — list, profile, dashboard, report, API

Tooltip

A tooltip that lies, lags, or hides off-screen breaks trust faster than a missing feature.



If you ship this — release is denied

- Tooltip cut off at the screen edge — does not auto-flip
- Vague text — ‘Click here for more information’
- Spelling errors in the tooltip body
- Disappears when pointer moves onto the tooltip body
- Esc does not dismiss it; hover-only (no keyboard trigger)
- No aria-describedby / role="tooltip" — screen readers ignore it



Pass these and submit with confidence

- Concise, specific text — answers the actual question
- Spelling and grammar reviewed before merge
- Auto-flips at screen edges; never cut off
- Persists when pointer moves onto the tooltip body
- Dismissible via Esc; keyboard-focusable, not just hover
- aria-describedby + role="tooltip"; respects prefers-reduced-motion

Links

Every broken link is a dead end. Every "click here" is a missed accessibility signal.



If you ship this — release is denied

- Generic 'click here' / 'more' text — no context for users or screen readers
- Link text indistinguishable from body text — same color, no underline
- External links use http:// — credentials leak in transit
- External links open in new tab without rel="noopener noreferrer"
- Multiple 404 / dead links present in production
- Anchor links target hidden behind sticky header



Pass these and submit with confidence

- Link text describes the destination — never 'click here' or bare URL
- Visually distinct: colored, underlined; hover and visited states differ
- External links use https:// only
- External target="_blank" links carry rel="noopener noreferrer"
- CI broken-link checker runs in pipeline; build fails on dead links
- mailto: / tel: links present where appropriate; visible focus ring on Tab

Upload control

File flows look fine in dev with small files and fast networks. Production breaks them.

✗ If you ship this — release is denied

- No file-type filter, no size limit, no progress
- Server accepts disallowed types (client-side check only)
- User clicks Upload twice — page appears frozen, file uploads twice
- Cancel button missing during upload; user cannot abort
- Generic 'Upload failed.' — no reason, no retry option
- 500 MB executable accepted as a profile picture

✓ Pass these and submit with confidence

- Allowed types and max size enforced on BOTH client AND server
- Drop-zone shows accepted formats and limits up front
- Progress bar with percentage during upload; cancel works
- Specific errors: 'File too large. Max is 10 MB.' / 'Type not supported.'
- Success state shows filename and size; ready for next step
- Upload button disabled while a request is in flight

Download / report (PDF, Excel)

When the report is wrong, the customer assumes the system is wrong. Every glitch is a credibility tax.

If you ship this — release is denied

- Title misspelled ('Reprot') on the cover
- Numbers shown without commas — '1000000' not '1,000,000'
- Content cut off at page bottom; sentences trail off mid-word
- No page numbers; cover page missing
- Filename is generic ('data.pdf'); not derived from report or upload
- Empty / no-data report still downloads as a blank PDF

Pass these and submit with confidence

- Logo, title, generated-on timestamp on cover and header
- Numbers consistently comma-formatted; currency symbol where relevant
- Page break logic clean — no orphan lines, no cut-off content
- Page numbers ('Page X of Y') on every page
- Filename derived from report or upload — never 'data.pdf'
- Empty state blocked with clear message; special characters render correctly

Header and navigation

The header runs across every screen. If it breaks on Safari, every screen is broken on Safari.



If you ship this — release is denied

- Header verified only on Chrome — broken layout on Safari and Edge
- Sticky header overlaps the first paragraph of body content
- Hamburger menu opens but does not close on outside click
- Broken images / missing icons in the header
- Logo does not link back to the home page
- Spelling errors in nav links ('Reprots', 'Setings')



Pass these and submit with confidence

- Header verified on Chrome, Firefox, Safari, Edge
- Sticky header has correct padding so content is never hidden
- Hamburger opens and closes on tap, link click, or outside click
- All header assets load; alt text on every icon
- Logo always navigates to the home page from any screen
- Active page underlined / colored to show current screen

Site footer

A broken footer signals a careless team. Spelling, alignment, links, copyright year — get all of them right.



If you ship this — release is denied

- Spelling errors throughout ('Cmpany', 'Carrers', 'Documentaion')
- Copyright year is stale ('© 2022' on a 2026 page)
- Some footer links return 404
- Social icons missing for required platforms
- Footer differs across pages — different links on Home vs Dashboard
- Footer breaks (overflows) on mobile — columns do not stack

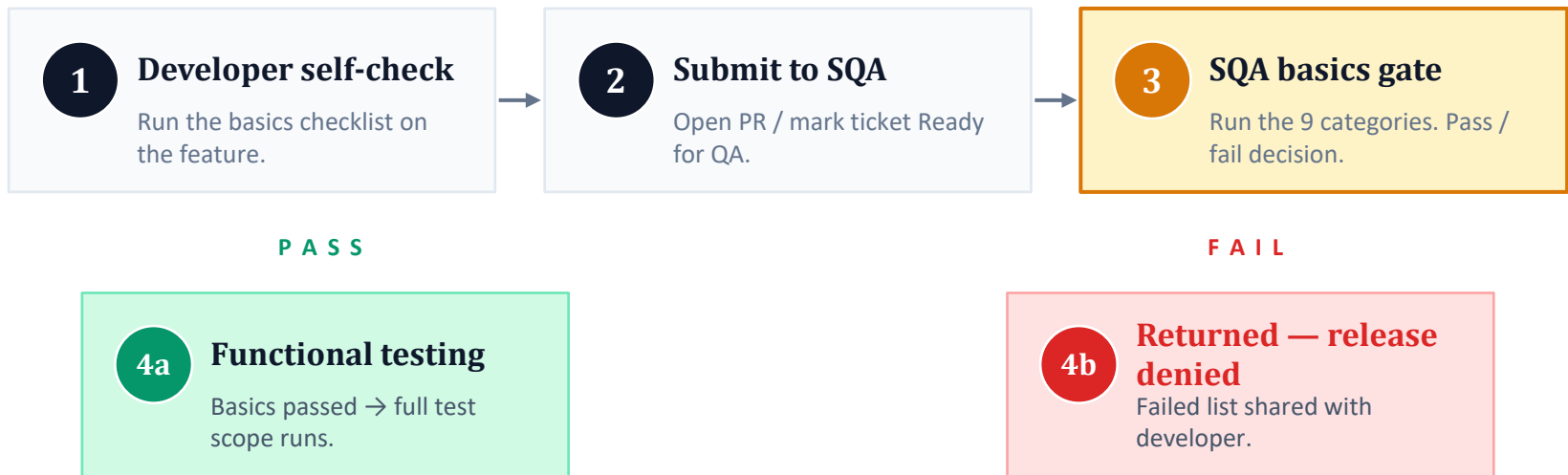


Pass these and submit with confidence

- Spelling and grammar reviewed before merge
- Copyright year auto-updates from server year — never stale
- All links resolve (HTTP 200/301/302) — verified by CI broken-link job
- Footer is identical across every page; logo links to home
- Newsletter email field validated; success and error messages clear
- Stacks cleanly on mobile and tablet — no overflow, no broken layout

How a build moves through SQA

Every build follows the same path. Failing the basics returns the build immediately — before any functional testing begins.



Developer self-check — run before every PR

One line per category. If you can't tick all 25, fix first — don't submit.

- ✓ Register: Complexity rules; CAPTCHA; email verify; alias-safe
- ✓ Login: Submit disabled until valid; masked password; rate-limit
- ✓ Email: Format validated; whitespace trimmed; case-normalized
- ✓ Password: Masked; eye toggle; strength meter; never logged
- ✓ OTP: 6 digits; HTTPS body; expires; single-use; rate-limited
- ✓ Captcha: Server-verified; single-use token; audio fallback
- ✓ Forgot pwd: HTTPS link; single-use; expires; no enumeration
- ✓ Buttons: Disabled while loading; double-click safe; keyboard OK
- ✓ Delete: Confirm modal; soft-delete + undo; RBAC; audit log
- ✓ Errors: Field-specific; spelling reviewed; consistent tone
- ✓ Input field: Trim; max-length; escape HTML; param SQL; utf8mb4
- ✓ Mobile: Digits only; country code prefix; stored in E.164
- ✓ Search: Icon + placeholder; Enter triggers; XSS / SQLi sanitized
- ✓ Dropdown: All values; correct order; long values ellipsis + tooltip
- ✓ Checkbox: Label clicks toggle; Space toggles; focus ring; 44px
- ✓ Radio: Same name attr; arrow keys switch; selection retained
- ✓ URL: Protocol enforced; whitespace trimmed; javascript: rejected
- ✓ Text area: HTML escaped; SQL param; max-length + counter
- ✓ Currency: Live commas; one decimal; max enforced; raw in DB
- ✓ Tooltip: Specific text; auto-flips; Esc dismisses; keyboard OK
- ✓ Links: Descriptive text; HTTPS; rel=noopener; CI 404 check
- ✓ Upload: Type + size enforced client AND server; progress + cancel
- ✓ Download: Page numbers; commas; filename derived; non-empty
- ✓ Header: Verified Chrome / Firefox / Safari / Edge; logo links home
- ✓ Footer: Identical across pages; auto-updating year; responsive

Prevention is faster than fixing. Twenty-five ticks here saves the whole team a cycle.